usb_power

File: usb_power.kicad_sch

USB−C sink (no host), VBUS protection, USB0 device/FEL

power

File: power.kicad_sch

Regulators, rail order, supervisors, staged−power links

f133

File: f133.kicad_sch

F133−B SoC: supplies, clock, reset, pin assignment

boot_nor

File: boot_nor.kicad_sch

32 MiB SPI NOR boot (BOOT−SEL 00), straps, FEL/clip recovery

mn820

File: mn820.kicad_sch

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel

t35

File: t35.kicad_sch

T35F400C3 (optional accelerator): rails, config flash, SPI0 isolation, mailbox

psram

File: psram.kicad_sch

APS6404L 8 MiB PSRAM: switched VDD, switched start−up pull−downs

sd_audio

File: sd_audio.kicad_sch

microSD game media (not boot), 2x MAX98357A I2S speakers

controls

File: controls.kicad_sch

Game buttons, LEDs, debug UART

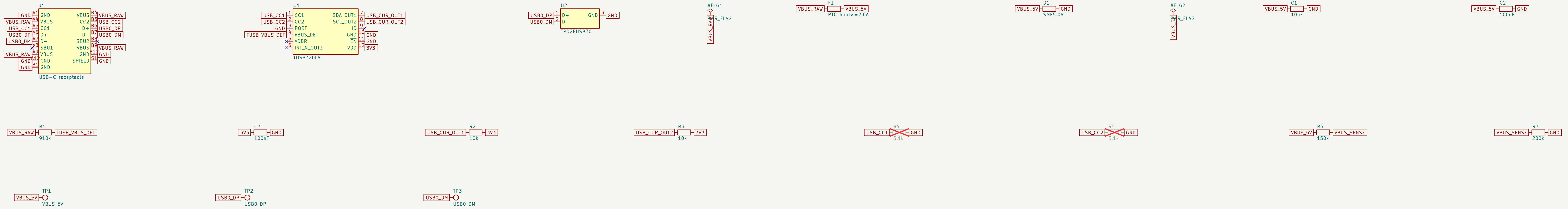
NOT approved for fabrication. LCD/panel/backlight deferred to R09.

Sheet: /
File: ARC1_R08.kicad_sch

Title: ARC−1 Compact R08 − hierarchical root

Size: A3
Date: 2026−10−15
Rev: R08−draft
KiCad E.D.A. 9.0.9
Id: 1/10

USB-C sink (no host), VBUS protection, USB0 device/FEL
Sheet generated from design.py; global labels carry every net.


J1
GND A1 GND
VBUS_RAW A4 VBUS
USB_CC1 A5 CC1
USB0_DP A6 D+
USB0_DM A7 D−
A8 SBU1
VBUS_RAW A9 VBUS
GND A12 GND
GND B1 GND
VBUS B4 VBUS_RAW
CC2 B5 USB_CC2
D+ B6 USB0_DP
D− B7 USB0_DM
SBU2 B8
VBUS B9 VBUS_RAW
GND B12 GND
SHIELD S1 GND
USB−C receptacle
U1
USB_CC1 1 CC1
USB_CC2 2 CC2
GND 3 PORT
TUSB_VBUS_DET 4 VBUS_DET
5 ADDR
6 INT_N_OUT3
SDA_OUT1 7 USB_CUR_OUT1
SCL_OUT2 8 USB_CUR_OUT2
ID 9
GND 10 GND
EN 11 GND
VDD 12 3V3
TUSB320LAI
U2
USB0_DP 1 D+
USB0_DM 2 D−
GND 3 GND
TPD2EUSB30
#FLG1
PWR_FLAG
VBUS_RAW
F1
VBUS_RAW
VBUS_5V
PTC hold>=2.6A
D1
VBUS_5V
GND
SMF5.0A
#FLG2
PWR_FLAG
VBUS_5V
C1
VBUS_5V
GND
10uF
C2
VBUS_5V
GND
100nF
R1
VBUS_RAW
TUSB_VBUS_DET
910k
C3
3V3
GND
100nF
R2
USB_CUR_OUT1
3V3
10k
R3
USB_CUR_OUT2
3V3
10k
R4
USB_CC1
GND
5.1k
R5
USB_CC2
GND
5.1k
R6
VBUS_5V
VBUS_SENSE
150k
R7
VBUS_SENSE
GND
200k
TP1
VBUS_5V
VBUS_5V
TP2
USB0_DP
USB0_DP
TP3
USB0_DM
USB0_DM

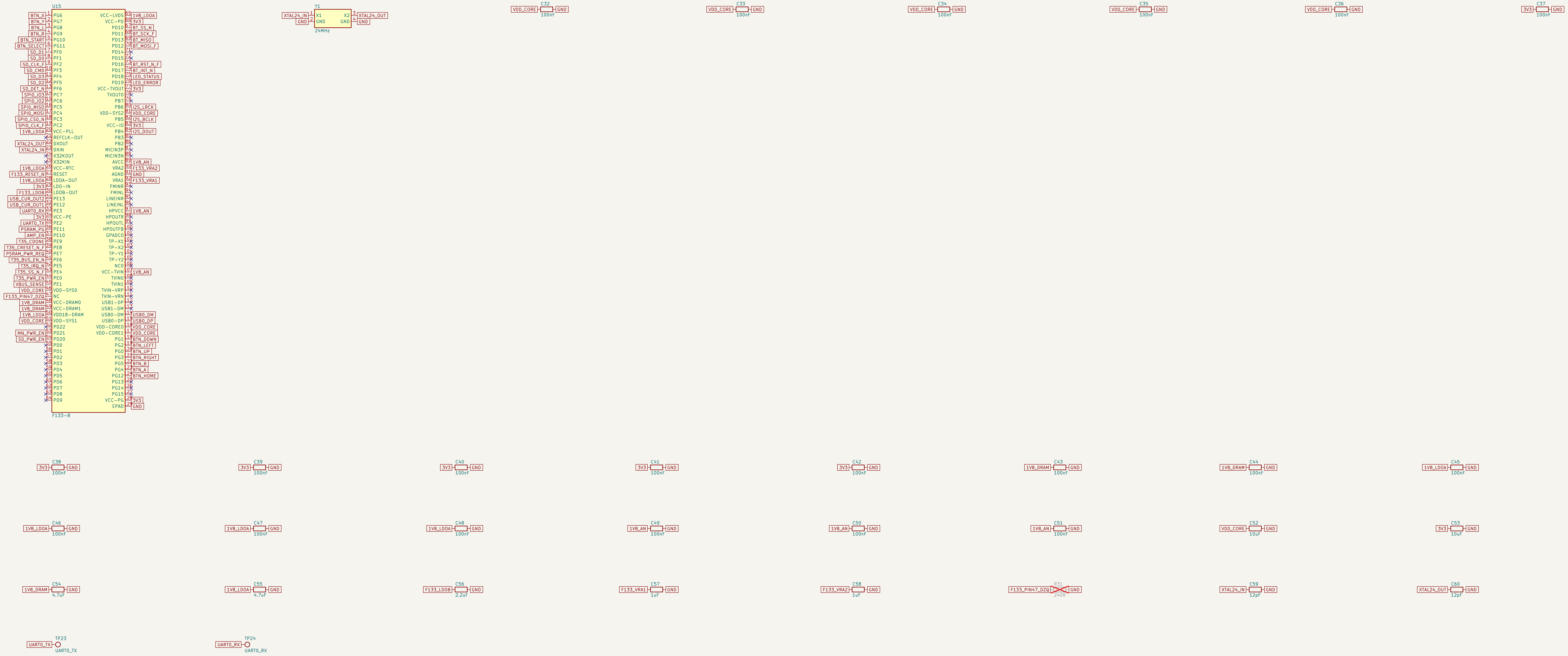
Generated by R08_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC−1
Sheet: /usb_power/
File: usb_power.kicad_sch
Title: ARC−1 Compact R08: USB−C sink (no host), VBUS protection, USB0 de
Size: User
Date: 2026−10−15
Rev: R08−draft
KiCad E.D.A. 9.0.9
Id: 2/10

Regulators, rail order, supervisors, staged−power links
Sheet generated from design.py; global labels carry every net.

U3
VBUS_5V 1 EN
GND 2 GND
SW_3V3 3 SW
VIN 4 VBUS_5V
PG 5 PG_3V3_BUCK
FB 6 FB_3V3
3V3 2A
U4
PG_3V3 1 RESET
GND 2 GND
SUP33_MR_N 3 MR
CT 4 SUP33_CT
SENSE 5 3V3
VDD 6 3V3
TPS3808G33
U5
PG_3V3 1 EN
GND 2 GND
SW_CORE 3 SW
VIN 4 VBUS_5V
FB 5 FB_CORE
VDD_CORE 0.95V
U6
PG_3V3 1 EN
GND 2 GND
SW_DRAM 3 SW
VIN 4 VBUS_5V
FB 5 FB_DRAM
1V8_DRAM
U7
3V3 1 IN
GND 2 GND
3V3 3 EN
NC 4
OUT 5 1V8_AN
1V8_AN
U8
F133_RESET_N 1 RESET
GND 2 GND
RST_BTN_N 3 MR
CT 4 SUPC_CT
SENSE 5 VDD_CORE
VDD 6 3V3
TPS3808G09
U9
T35_PWR_EN 1 EN
GND 2 GND
SW_1V2 3 SW
VIN 4 VBUS_5V
PG 5 T35_1V2_PG
FB 6 FB_1V2
1V2_T35
U10
3V3 1 VIN
GND 2 GND
T35_1V2_PG 3 ON
CT 4 SW_T35_CT
QOD 5 3V3_T35
VOUT 6 3V3_T35
3V3_T35 switch
U11
3V3 1 VIN
GND 2 GND
MN_PWR_EN 3 ON
CT 4 SW_MN_CT
QOD 5 3V3_MN
VOUT 6 3V3_MN
3V3_MN switch
U12
3V3_T35 1 VIN
GND 2 GND
PSRAM_EN 3 ON
CT 4 SW_PS_CT
QOD 5 PSRAM_VDD
VOUT 6 PSRAM_VDD
PSRAM_VDD switch
U13
PSRAM_PWR_REQ 1 A
CDONE_DLY 2 B
GND 3 GND
Y 4 PSRAM_EN
VCC 5 3V3
AND
U14
PSRAM_PG 1 RESET
GND 2 GND
PSRAM_SUP_MR_N 3 MR
CT 4 PSRAM_SUP_CT
SENSE 5 PSRAM_VDD
VDD 6 3V3_T35
TPS3808G30
SW1
RST_BTN_N 1 1
2 2 GND
RESET
C4 VBUS_5V GND 10uF
L1 SW_3V3 3V3_REG 2.2uH
R8 3V3_REG FB_3V3 453k
R9 FB_3V3 GND 100k
C5 3V3_REG GND 22uF
C6 3V3_REG GND 22uF
R10 PG_3V3_BUCK 3V3_REG 100k
#FLG3 PWR_FLAG 3V3_REG
R11 3V3_REG 3V3 0R
#FLG4 PWR_FLAG 3V3
TP4 3V3_REG 3V3_REG
TP5 3V3 3V3
TP6 PG_3V3_BUCK PG_3V3_BUCK
C7 SUP33_CT GND 2.2nF
R12 SUP33_MR_N 3V3 100k
R13 PG_3V3 3V3 10k
C8 3V3 GND 100nF
C9 VBUS_5V GND 10uF
L2 SW_CORE VDD_CORE_REG 2.2uH
R14 VDD_CORE_REG FB_CORE 57.6k
R15 FB_CORE GND 100k
C10 VDD_CORE_REG GND 22uF
C11 VDD_CORE_REG GND 22uF
#FLG5 PWR_FLAG VDD_CORE_REG
R16 VDD_CORE_REG VDD_CORE 0R
#FLG6 PWR_FLAG VDD_CORE
TP7 VDD_CORE VDD_CORE
C12 VBUS_5V GND 10uF
L3 SW_DRAM 1V8_DRAM 2.2uH
R17 1V8_DRAM FB_DRAM 200k
R18 FB_DRAM GND 100k
C13 1V8_DRAM GND 22uF
#FLG7 PWR_FLAG 1V8_DRAM
TP8 1V8_DRAM 1V8_DRAM
C14 3V3 GND 1uF
C15 1V8_AN GND 1uF
TP9 1V8_AN 1V8_AN
C16 SUPC_CT GND 15nF
R19 RST_BTN_N 3V3 100k
R20 F133_RESET_N 1V8_LDOA 10k
TP10 F133_RESET_N F133_RESET_N
TP11 PG_3V3 PG_3V3
C17 VBUS_5V GND 10uF
L4 SW_1V2 1V2_T35_REG 2.2uH
R21 1V2_T35_REG FB_1V2 100k
R22 FB_1V2 GND 100k
C18 1V2_T35_REG GND 22uF
C19 1V2_T35_REG GND 22uF
#FLG8 PWR_FLAG 1V2_T35_REG
R23 1V2_T35_REG 1V2_T35 0R
#FLG9 PWR_FLAG 1V2_T35
R24 T35_PWR_EN GND 100k
R25 T35_1V2_PG 3V3 100k
TP12 1V2_T35 1V2_T35
TP13 T35_1V2_PG T35_1V2_PG
C20 SW_T35_CT GND 1nF
C21 3V3 GND 1uF
C22 3V3_T35 GND 10uF
TP14 3V3_T35 3V3_T35
C23 SW_MN_CT GND 1nF
C24 3V3 GND 1uF
C25 3V3_MN GND 22uF
R26 MN_PWR_EN 3V3 100k
TP15 3V3_MN 3V3_MN
C26 SW_PS_CT GND 1nF
C27 3V3_T35 GND 1uF
C28 PSRAM_VDD GND 4.7uF
C29 3V3 GND 100nF
R27 PSRAM_PWR_REQ GND 100k
R28 PSRAM_EN GND 100k
C30 PSRAM_SUP_CT GND 1nF
R29 PSRAM_SUP_MR_N 3V3_T35 100k
R30 PSRAM_PG 3V3_T35 10k
C31 3V3_T35 GND 100nF
TP16 PSRAM_VDD PSRAM_VDD
TP17 PSRAM_PG PSRAM_PG
TP18 PSRAM_EN PSRAM_EN
TP19 GND GND
TP20 GND GND
TP21 GND GND
TP22 GND GND
#FLG10 PWR_FLAG GND
FID1 fiducial
FID2 fiducial
FID3 fiducial
H1 GND M2.5
H2 GND M2.5
H3 GND M2.5
H4 GND M2.5

Generated by R08_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC−1
Sheet: /power/
File: power.kicad_sch
Title: ARC−1 Compact R08: Regulators, rail order, supervisors, staged−power
Size: User
Date: 2026−10−15
Rev: R08−draft
KiCad E.D.A. 9.0.9
Id: 3/10

F133-B SoC: supplies, clock, reset, pin assignment
Sheet generated from design.py; global labels carry every net.


U15
F133-B
BTN_X 1 PG6
BTN_Y 2 PG7
BTN_L 3 PG8
BTN_R 4 PG9
BTN_START 5 PG10
BTN_SELECT 6 PG11
SD_D1 7 PF0
SD_D0 8 PF1
SD_CLK_F 9 PF2
SD_CMD 10 PF3
SD_D3 11 PF4
SD_D2 12 PF5
SD_DET_N 13 PF6
SPI0_IO3 14 PC7
SPI0_IO2 15 PC6
SPI0_MISO 16 PC5
SPI0_MOSI 17 PC4
SPI0_CS0_N 18 PC3
SPI0_CLK_F 19 PC2
1V8_LDOA 20 VCC-PLL
21 REFCLK-OUT
XTAL24_OUT 22 DXOUT
XTAL24_IN 23 DXIN
24 X32KOUT
25 X32KIN
1V8_LDOA 26 VCC-RTC
F133_RESET_N 27 RESET
1V8_LDOA 28 LDOA-OUT
3V3 29 LDO-IN
F133_LDOB 30 LDOB-OUT
USB_CUR_OUT2 31 PE13
USB_CUR_OUT1 32 PE12
UART0_RX 33 PE3
3V3 34 VCC-PE
UART0_TX 35 PE2
PSRAM_PG 36 PE11
AMP_EN 37 PE10
T35_CDONE 38 PE9
T35_CRESET_N_F 39 PE8
PSRAM_PWR_REQ 40 PE7
T35_BUS_EN_N 41 PE6
T35_IRQ_N 42 PE5
T35_SS_N_F 43 PE4
T35_PWR_EN 44 PE0
VBUS_SENSE 45 PE1
VDD_CORE 46 VDD-SYS0
F133_PIN47_DZQ 47 NC
1V8_DRAM 48 VCC-DRAM0
1V8_DRAM 49 VCC-DRAM1
1V8_LDOA 50 VDD18-DRAM
VDD_CORE 51 VDD-SYS1
52 PD22
MN_PWR_EN 53 PD21
SD_PWR_EN 54 PD20
55 PD0
56 PD1
57 PD2
58 PD3
59 PD4
60 PD5
61 PD6
62 PD7
63 PD8
64 PD9
VCC-LVDS 65 1V8_LDOA
VCC-PD 66 3V3
PD10 67 BT_SS_N
PD11 68 BT_SCK_F
PD13 69 BT_MISO
PD12 70 BT_MOSI_F
PD14 71
PD15 72
PD16 73 BT_RST_N_F
PD17 74 BT_INT_N
PD18 75 LED_STATUS
PD19 76 LED_ERROR
VCC-TVOUT 77 3V3
TVOUT0 78
PB7 79
PB6 80 I2S_LRCK
VDD-SYS2 81 VDD_CORE
PB5 82 I2S_BCLK
VCC-IO 83 3V3
PB4 84 I2S_DOUT
PB3 85
PB2 86
MICIN3P 87
MICIN3N 88
AVCC 89 1V8_AN
VRA2 90 F133_VRA2
AGND 91 GND
VRA1 92 F133_VRA1
FMINR 93
FMINL 94
LINEINR 95
LINEINL 96
HPVCC 97 1V8_AN
HPOUTR 98
HPOUTL 99
HPOUTFB 100
GPADC0 101
TP-X1 102
TP-X2 103
TP-Y1 104
TP-Y2 105
NC0 106
VCC-TVIN 107 1V8_AN
TVIN0 108
TVIN1 109
TVIN-VRP 110
TVIN-VRN 111
USB1-DP 112
USB1-DM 113
USB0-DM 114 USB0_DM
USB0-DP 115 USB0_DP
VDD-CORE0 116 VDD_CORE
VDD-CORE1 117 VDD_CORE
PG1 118 BTN_DOWN
PG2 119 BTN_LEFT
PG0 120 BTN_UP
PG3 121 BTN_RIGHT
PG5 122 BTN_B
PG4 123 BTN_A
PG12 124 BTN_HOME
PG13 125
PG14 126
PG15 127
VCC-PG 128 3V3
EPAD 129 GND
Y1
XTAL24_IN 1 X1
GND 2 GND
X2 3 XTAL24_OUT
GND 4 GND
24MHz
VDD_CORE C32 100nF GND
VDD_CORE C33 100nF GND
VDD_CORE C34 100nF GND
VDD_CORE C35 100nF GND
VDD_CORE C36 100nF GND
3V3 C37 100nF GND
3V3 C38 100nF GND
3V3 C39 100nF GND
3V3 C40 100nF GND
3V3 C41 100nF GND
3V3 C42 100nF GND
1V8_DRAM C43 100nF GND
1V8_DRAM C44 100nF GND
1V8_LDOA C45 100nF GND
1V8_LDOA C46 100nF GND
1V8_LDOA C47 100nF GND
1V8_LDOA C48 100nF GND
1V8_AN C49 100nF GND
1V8_AN C50 100nF GND
1V8_AN C51 100nF GND
VDD_CORE C52 10uF GND
3V3 C53 10uF GND
1V8_DRAM C54 4.7uF GND
1V8_LDOA C55 4.7uF GND
F133_LDOB C56 2.2uF GND
F133_VRA1 C57 1uF GND
F133_VRA2 C58 1uF GND
F133_PIN47_DZQ R31 240R GND
XTAL24_IN C59 12pF GND
XTAL24_OUT C60 12pF GND
UART0_TX TP23 UART0_TX
UART0_RX TP24 UART0_RX

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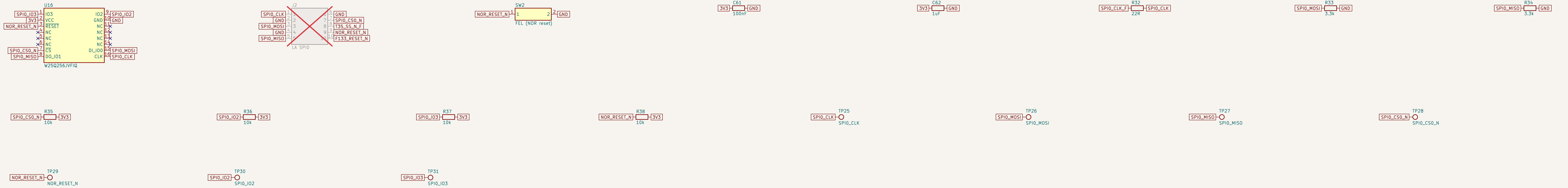
STRUTHIO ARC-1

Sheet: /f133/
File: f133.kicad_sch

Title: ARC-1 Compact R08: F133-B SoC: supplies, clock, reset, pin assignm

Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 4/10

32 MiB SPI NOR boot (BOOT−SEL 00), straps, FEL/clip recovery
Sheet generated from design.py; global labels carry every net.


U16
W25Q256JVFIQ
1 IO3 SPI0_IO3
2 VCC 3V3
3 RESET NOR_RESET_N
4 NC
5 NC
6 NC
7 CS SPI0_CS0_N
8 DO_IO1 SPI0_MISO
9 IO2 SPI0_IO2
10 GND GND
11 NC
12 NC
13 NC
14 NC
15 DI_IO0 SPI0_MOSI
16 CLK SPI0_CLK
J2
LA SPI0
1 SPI0_CLK
2 GND
3 SPI0_MOSI
4 GND
5 SPI0_MISO
6 GND
7 SPI0_CS0_N
8 T35_SS_N_F
9 NOR_RESET_N
10 F133_RESET_N
SW2
FEL (NOR reset)
NOR_RESET_N 1
2 GND
C61 100nF 3V3 GND
C62 1uF 3V3 GND
R32 22R SPI0_CLK_F SPI0_CLK
R33 3.3k SPI0_MOSI GND
R34 3.3k SPI0_MISO GND
R35 10k SPI0_CS0_N 3V3
R36 10k SPI0_IO2 3V3
R37 10k SPI0_IO3 3V3
R38 10k NOR_RESET_N 3V3
TP25 SPI0_CLK
TP26 SPI0_MOSI
TP27 SPI0_MISO
TP28 SPI0_CS0_N
TP29 NOR_RESET_N
TP30 SPI0_IO2
TP31 SPI0_IO3

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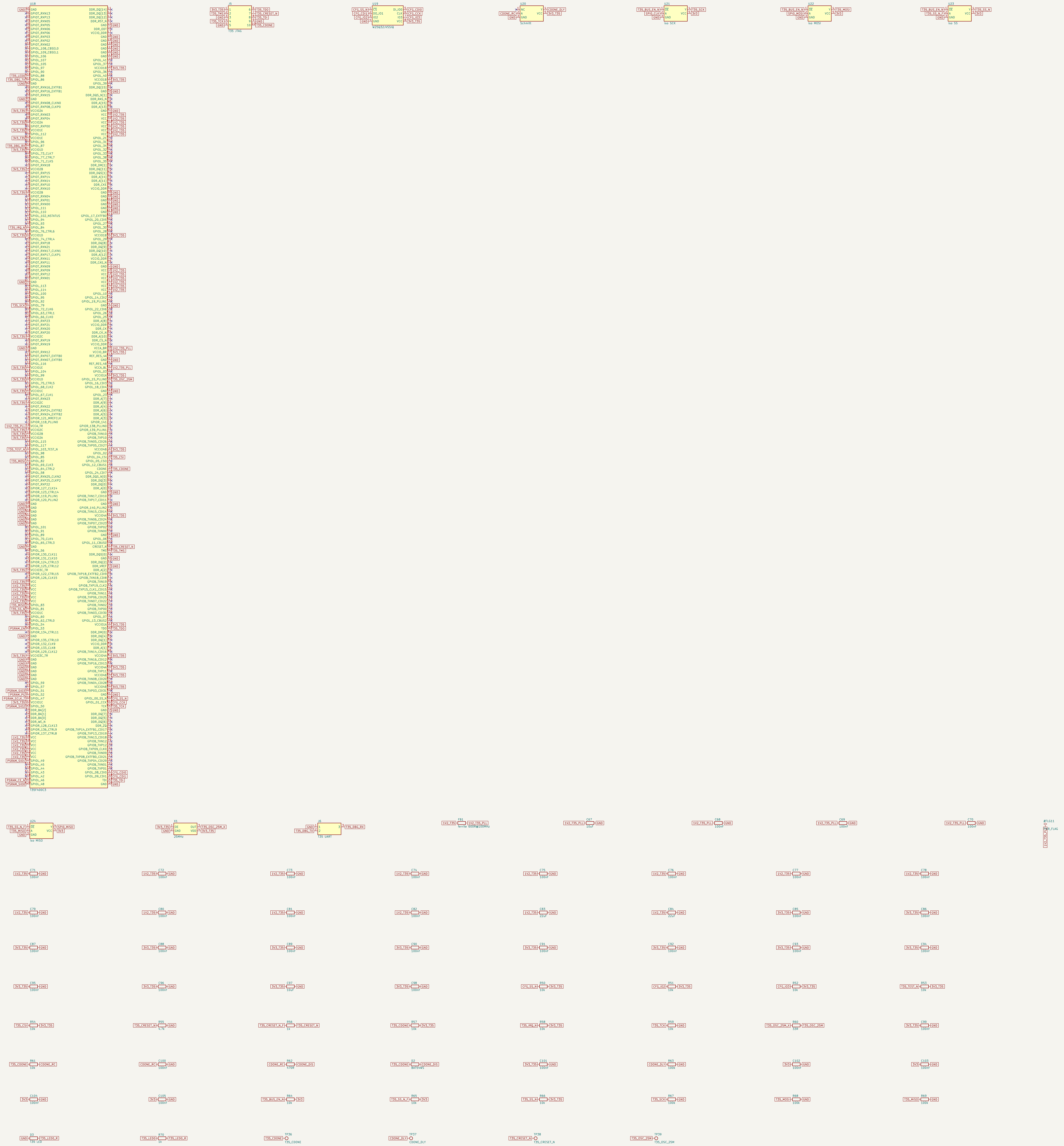
SELECTED_UNVERIFIED engineering capture − NOT released for fabrication
STRUTHIO ARC−1
Sheet: /boot_nor/
File: boot_nor.kicad_sch
Title: ARC−1 Compact R08: 32 MiB SPI NOR boot (BOOT−SEL 00), straps, FE
Size: User
Date: 2026−10−15
Rev: R08−draft
KiCad E.D.A. 9.0.9
Id: 5/10

MN820 (BT820B + 128 MiB DDR3L) module socket, host SPI1, LVDS to R09 panel
Sheet generated from design.py; global labels carry every net.

J3
1 GND GND
3 GND GND
5 NC
7 NC
9 NC
11 NC
13 VCC3V3 3V3_MN
15 VCC3V3 3V3_MN
17 VCC3V3 3V3_MN
19 VCC3V3 3V3_MN
21 SD_CD
23 SD_D0
25 SD_D1
27 SD_D2
29 SD_D3
31 SD_CLK
33 SD_CMD
35 SPIM_MOSI
37 SPIM_MISO
39 SPIM_IO3
41 SPIM_IO2
43 SPIM_SS_N
45 SPIM_SCLK
47 GND GND
49 GND GND
51 GND GND
53 GND GND
55 GND GND
57 LVDS_TX0_D0_N
59 LVDS_TX0_D0_P
61 LVDS_TX0_D1_N
63 LVDS_TX0_D1_P
65 LVDS_TX0_D2_N
67 LVDS_TX0_D2_P
69 LVDS_TX0_CLK_N
71 LVDS_TX0_CLK_P
73 LVDS_TX0_D3_N
75 LVDS_TX0_D3_P
77 GND GND
79 LVDS_TX1_D0_N
81 LVDS_TX1_D0_P
83 LVDS_TX1_D1_N
85 LVDS_TX1_D1_P
87 LVDS_TX1_D2_N
89 LVDS_TX1_D2_P
91 LVDS_TX1_CLK_N
93 LVDS_TX1_CLK_P
95 LVDS_TX1_D3_N
97 LVDS_TX1_D3_P
99 GND GND
2 GND GND
4 LVDS_RX0_D0_N
6 LVDS_RX0_D0_P
8 LVDS_RX0_D1_N
10 LVDS_RX0_D1_P
12 LVDS_RX0_D2_N
14 LVDS_RX0_D2_P
16 LVDS_RX0_CLK_N
18 LVDS_RX0_CLK_P
20 LVDS_RX0_D3_N
22 LVDS_RX0_D3_P
24 GND GND
26 LVDS_RX1_D0_N
28 LVDS_RX1_D0_P
30 LVDS_RX1_D1_N
32 LVDS_RX1_D1_P
34 LVDS_RX1_D2_N
36 LVDS_RX1_D2_P
38 LVDS_RX1_CLK_N
40 LVDS_RX1_CLK_P
42 LVDS_RX1_D3_N
44 LVDS_RX1_D3_P
46 GND GND
48 AUDIO_R
50 GND GND
52 AUDIO_L
54 GND GND
56 I2S_LRCLK
58 I2S_BCLK
60 I2S_SDA
62 DISP
64 BKLIT_PWM
66 GND GND
68 GPIO3
70 GPIO2
72 GPIO4
74 GPIO6
76 GPIO5
78 GPIO7
80 GPIO8
82 GND GND
84 SPIS_MISO BT_MISO_M
86 SPIS_SCLK BT_SCK
88 SPIS_SS_N BT_SS_N_M
90 SPIS_MOSI BT_MOSI
92 SPIS_IO3 BT_IO3_M
94 SPIS_IO2 BT_IO2_M
96 INT_N BT_INT_N
98 RESET_N BT_RST_N
100 GND GND
DF40C-100DS-0.4V(51)
J4
BT_SCK 1
GND 2
BT_MOSI 3
GND 4
BT_MISO 5
6 GND
7 BT_SS_N
8 BT_INT_N
9 BT_RST_N
10 GND
LA SPI1
U17
BT_SS_N 1 $\overline{OE}$
BT_MISO_M 2 A
GND 3 GND
Y 4 BT_MISO
VCC 5 3V3
MISO buffer
BT_SCK_F R39 33R BT_SCK
BT_MOSI_F R40 33R BT_MOSI
BT_SS_N R41 33R BT_SS_N_M
BT_RST_N_F R42 33R BT_RST_N
BT_SS_N_M R43 10k 3V3_MN
BT_INT_N R44 10k 3V3_MN
BT_RST_N R45 10k GND
BT_MISO_M R46 10k GND
BT_IO2_M R47 10k GND
BT_IO3_M R48 10k GND
BT_MISO R49 47k GND
3V3 C63 100nF GND
3V3_MN C64 100nF GND
3V3_MN C65 100nF GND
3V3_MN C66 10uF GND
H5 MN820 hole
H6 MN820 hole
BT_SCK TP32 BT_SCK
BT_MISO TP33 BT_MISO
BT_INT_N TP34 BT_INT_N
BT_RST_N TP35 BT_RST_N

Generated by R08_WORK/cad/gen/gen_kicad.py from design.py — edit the source, not this file
SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1
Sheet: /mn820/
File: mn820.kicad_sch
Title: ARC-1 Compact R08: MN820 (BT820B + 128 MiB DDR3L) module sock
Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 6/10

T35F400C3 (optional accelerator): rails, config flash, SPI0 isolation, mailbox
Sheet generated from design.py; global labels carry every net.


U18
T35F400C3
J5
T35 JTAG
3V3_T35
T35_TMS
GND
T35_TCK
T35_TDO
T35_CRESET_N
T35_TDI
T35_CDONE
U19
W25Q32JVSSIQ
CFG_SS_N
CFG_CDI1
CFG_IO2
CFG_CDI0
CFG_CCK
CFG_IO3
U20
Schmitt
CDONE_RC
CDONE_DLY
U21
iso SCK
T35_BUS_EN_N
SPI0_CLK
T35_SCK
3V3
U22
iso MOSI
SPI0_MOSI
T35_MOSI
U23
iso SS
T35_SS_N_F
T35_SS_N
U24
iso MISO
T35_MISO
SPI0_MISO
X1
25MHz
T35_OSC_25M_X
J6
T35 UART
T35_DBG_TX
T35_DBG_RX
FB1
ferrite 600R@100MHz
1V2_T35
1V2_T35_PLL
C67 10uF
C68 100nF
C69 100nF
C70 100nF
#FLG11
PWR_FLAG
C71 100nF
C72 100nF
C73 100nF
C74 100nF
C75 100nF
C76 100nF
C77 100nF
C78 100nF
C79 100nF
C80 100nF
C81 100nF
C82 100nF
C83 22uF
C84 22uF
C85 100nF
C86 100nF
C87 100nF
C88 100nF
C89 100nF
C90 100nF
C91 100nF
C92 100nF
C93 100nF
C94 100nF
C95 100nF
C96 100nF
C97 10uF
C98 100nF
R50 10k
R51 10k
R52 10k
R53 10k
T35_TEST_N
R54 10k
T35_CSI
R55 4.7k
R56 1k
T35_CRESET_N_F
R57 10k
R58 10k
T35_IRQ_N
R59 10k
R60 33R
T35_OSC_25M
C99 100nF
R61 10k
C100 100nF
R62 470R
CDONE_DIS
D2 BAT54WS
C101 100nF
R63 100k
C102 100nF
C103 100nF
C104 100nF
C105 100nF
R64 10k
R65 10k
R66 10k
R67 100k
R68 100k
R69 100k
D3
T35 LED
T35_LED0_R
R70 1k
T35_LED0
TP36
TP37
TP38
TP39

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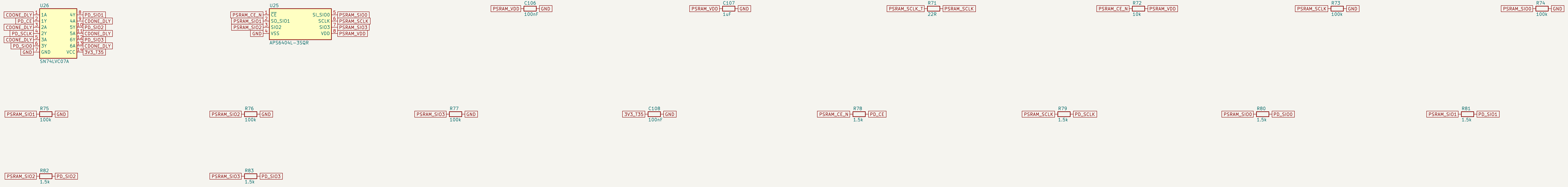
STRUTHIO ARC-1
Sheet: /t35/
File: t35.kicad_sch
Title: ARC-1 Compact R08: T35F400C3 (optional accelerator): rails, config f
Size: User | Date: 2026-10-15 | **Rev: R08-draft**
KiCad E.D.A. 9.0.9 | Id: 7/10

APS6404L 8 MiB PSRAM: switched VDD, switched start-up pull-downs
Sheet generated from design.py; global labels carry every net.


U26
CDONE_DLY 1 1A
PD_CE 2 1Y
CDONE_DLY 3 2A
PD_SCLK 4 2Y
CDONE_DLY 5 3A
PD_SIO0 6 3Y
GND 7 GND
4Y 8 PD_SIO1
4A 9 CDONE_DLY
5Y 10 PD_SIO2
5A 11 CDONE_DLY
6Y 12 PD_SIO3
6A 13 CDONE_DLY
VCC 14 3V3_T35
SN74LVC07A
U25
PSRAM_CE_N 1 CE
PSRAM_SIO1 2 SO_SIO1
PSRAM_SIO2 3 SIO2
GND 4 VSS
SI_SIO0 5 PSRAM_SIO0
SCLK 6 PSRAM_SCLK
SIO3 7 PSRAM_SIO3
VDD 8 PSRAM_VDD
APS6404L-3SQR
PSRAM_VDD C106 GND 100nF
PSRAM_VDD C107 GND 1uF
PSRAM_SCLK_T R71 PSRAM_SCLK 22R
PSRAM_CE_N R72 PSRAM_VDD 10k
PSRAM_SCLK R73 GND 100k
PSRAM_SIO0 R74 GND 100k
PSRAM_SIO1 R75 GND 100k
PSRAM_SIO2 R76 GND 100k
PSRAM_SIO3 R77 GND 100k
3V3_T35 C108 GND 100nF
PSRAM_CE_N R78 PD_CE 1.5k
PSRAM_SCLK R79 PD_SCLK 1.5k
PSRAM_SIO0 R80 PD_SIO0 1.5k
PSRAM_SIO1 R81 PD_SIO1 1.5k
PSRAM_SIO2 R82 PD_SIO2 1.5k
PSRAM_SIO3 R83 PD_SIO3 1.5k

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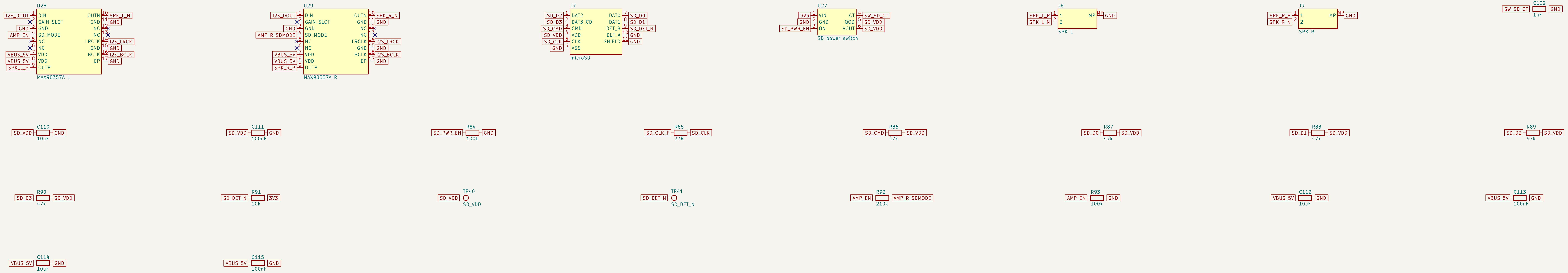
STRUTHIO ARC-1

Sheet: /psram/
File: psram.kicad_sch

Title: ARC-1 Compact R08: APS6404L 8 MiB PSRAM: switched VDD, switched

Size: User	Date: 2026-10-15	**Rev: R08-draft**
KiCad E.D.A. 9.0.9		Id: 8/10

microSD game media (not boot), 2x MAX98357A I2S speakers
Sheet generated from design.py; global labels carry every net.


U28
MAX98357A L
I2S_DOUT 1 DIN
2 GAIN_SLOT
GND 3 GND
AMP_EN 4 SD_MODE
5 NC
6 NC
VBUS_5V 7 VDD
VBUS_5V 8 VDD
SPK_L_P 9 OUTP
OUTN 10 SPK_L_N
GND 11 GND
NC 12
NC 13
LRCLK 14 I2S_LRCK
GND 15 GND
BCLK 16 I2S_BCLK
EP 17 GND
U29
MAX98357A R
I2S_DOUT 1 DIN
2 GAIN_SLOT
GND 3 GND
AMP_R_SDMODE 4 SD_MODE
5 NC
6 NC
VBUS_5V 7 VDD
VBUS_5V 8 VDD
SPK_R_P 9 OUTP
OUTN 10 SPK_R_N
GND 11 GND
NC 12
NC 13
LRCLK 14 I2S_LRCK
GND 15 GND
BCLK 16 I2S_BCLK
EP 17 GND
J7
microSD
SD_D2 1 DAT2
SD_D3 2 DAT3_CD
SD_CMD 3 CMD
SD_VDD 4 VDD
SD_CLK 5 CLK
GND 6 VSS
DAT0 7 SD_D0
DAT1 8 SD_D1
DET_B 9 SD_DET_N
DET_A 10 GND
SHIELD 11 GND
U27
SD power switch
3V3 1 VIN
GND 2 GND
SD_PWR_EN 3 ON
CT 4 SW_SD_CT
QOD 5 SD_VDD
VOUT 6 SD_VDD
J8
SPK L
SPK_L_P 1 1
SPK_L_N 2 2
MP MP GND
J9
SPK R
SPK_R_P 1 1
SPK_R_N 2 2
MP MP GND
C109 SW_SD_CT GND 1nF
C110 SD_VDD GND 10uF
C111 SD_VDD GND 100nF
R84 SD_PWR_EN GND 100k
R85 SD_CLK_F SD_CLK 33R
R86 SD_CMD SD_VDD 47k
R87 SD_D0 SD_VDD 47k
R88 SD_D1 SD_VDD 47k
R89 SD_D2 SD_VDD 47k
R90 SD_D3 SD_VDD 47k
R91 SD_DET_N 3V3 10k
TP40 SD_VDD SD_VDD
TP41 SD_DET_N SD_DET_N
R92 AMP_EN AMP_R_SDMODE 210k
R93 AMP_EN GND 100k
C112 VBUS_5V GND 10uF
C113 VBUS_5V GND 100nF
C114 VBUS_5V GND 10uF
C115 VBUS_5V GND 100nF

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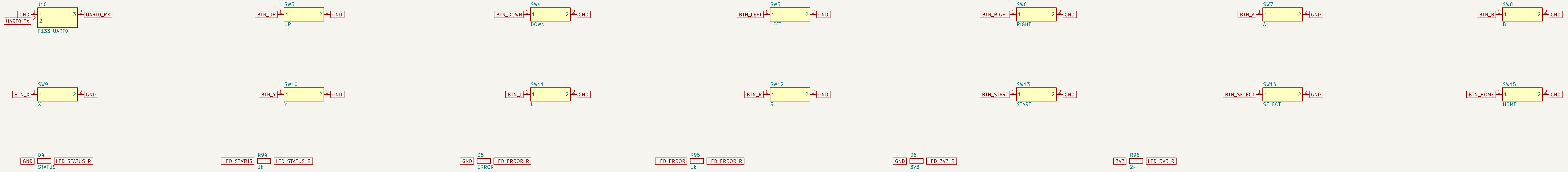
STRUTHIO ARC-1
Sheet: /sd_audio/
File: sd_audio.kicad_sch
Title: ARC-1 Compact R08: microSD game media (not boot), 2x MAX98357A
Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 9/10

Game buttons, LEDs, debug UART
Sheet generated from design.py: global labels carry every net.


J10
GND
1
1
UART0_TX
2
2
3
3
UART0_RX
F133 UART0
SW3
BTN_UP
1
1
2
2
GND
UP
SW4
BTN_DOWN
1
1
2
2
GND
DOWN
SW5
BTN_LEFT
1
1
2
2
GND
LEFT
SW6
BTN_RIGHT
1
1
2
2
GND
RIGHT
SW7
BTN_A
1
1
2
2
GND
A
SW8
BTN_B
1
1
2
2
GND
B
SW9
BTN_X
1
1
2
2
GND
X
SW10
BTN_Y
1
1
2
2
GND
Y
SW11
BTN_L
1
1
2
2
GND
L
SW12
BTN_R
1
1
2
2
GND
R
SW13
BTN_START
1
1
2
2
GND
START
SW14
BTN_SELECT
1
1
2
2
GND
SELECT
SW15
BTN_HOME
1
1
2
2
GND
HOME
D4
GND
LED_STATUS_R
STATUS
R94
LED_STATUS
LED_STATUS_R
1k
D5
GND
LED_ERROR_R
ERROR
R95
LED_ERROR
LED_ERROR_R
1k
D6
GND
LED_3V3_R
3V3
R96
3V3
LED_3V3_R
2k

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SELECTED_UNVERIFIED engineering capture — NOT released for fabrication
STRUTHIO ARC-1

Sheet: /controls/
File: controls.kicad_sch

Title: ARC-1 Compact R08: Game buttons, LEDs, debug UART

Size: User
Date: 2026-10-15
Rev: R08-draft
KiCad E.D.A. 9.0.9
Id: 10/10